

FinanceTools Pro - Deployment Guide

Project Structure

```
financetools-pro/
├─ index.html      # Main frontend (single-file SPA)
├─ server.js       # Node.js backend proxy for DeepSeek AI
├─ package.json    # Dependencies
├─ .env.example    # Environment template
├─ .gitignore      # Git ignore rules
├─ robots.txt     # SEO crawler instructions
├─ sitemap.xml     # Search engine sitemap
└─ DEPLOYMENT.md  # This file
```

Quick Start (Local Development)

Option A: Static Only (No AI Chat)

Simply open index.html in any browser or serve with any static server:

```
# Python 3
python -m http.server 8000

# Node.js (npx)
npx serve .

# PHP
php -S localhost:8000
```

Option B: Full Stack (With AI Chat)

```
# 1. Install dependencies
npm install

# 2. Create environment file
cp .env.example .env

# 3. Edit .env and add your DeepSeek API key
# DEEPSEEK_API_KEY=sk-05bc8a86ded1487b9930f9637d6354b2

# 4. Start server
npm start

# Server runs at http://localhost:3000
```

Production Deployment

1. Netlify (Recommended for Frontend)

Method A: Drag & Drop

1. Go to netlify.com
2. Drag the project folder to deploy
3. Set custom domain in Site Settings
4. Add `_redirects` file for SPA routing: `/* /index.html 200`

Method B: Git Integration

1. Push to GitHub/GitLab
2. Connect repo to Netlify
3. Build command: `echo "Static site"`
4. Publish directory: `/`

Netlify Settings:

- Enable HTTPS (auto-SSL)
- Enable asset optimization
- Set Cache-Control: `public, max-age=31536000` for static assets

2. Vercel (Recommended for Full Stack)

```
# Install Vercel CLI
npm i -g vercel

# Deploy
vercel

# Set environment variables in dashboard:
# DEEPSEEK_API_KEY=your_key
# ALLOWED_ORIGINS=https://yourdomain.com
```

vercel.json configuration:

```
{
  "version": 2,
  "builds": [
    { "src": "server.js", "use": "@vercel/node" }
  ],
  "routes": [
    { "src": "/api/(.*)", "dest": "/server.js" },
    { "src": "/(.*)", "dest": "/index.html" }
  ]
}
```

3. Cloudflare Pages (Best for Speed)

1. Connect GitHub repo
2. Build settings:
 - Framework preset: None
 - Build command: None
 - Output directory: /
3. Add `_routes.json` for Functions:

```
{
  "version": 1,
  "include": ["/api/*"],
  "exclude": ["/*"]
}
```

4. Traditional VPS (DigitalOcean, Linode, AWS EC2)

```
# On your server:
git clone <your-repo>
cd financetools-pro
npm install

# Create .env file with production values
nano .env

# Install PM2 for process management
npm install -g pm2

# Start with PM2
pm2 start server.js --name "financetools"
pm2 startup
pm2 save

# Configure Nginx reverse proxy
sudo nano /etc/nginx/sites-available/financetools
```

Nginx Configuration:

```
server {
    listen 80;
    server_name yourdomain.com www.yourdomain.com;
    return 301 https://$server_name$request_uri;
}

server {
    listen 443 ssl http2;
    server_name yourdomain.com;

    ssl_certificate /path/to/cert.pem;
    ssl_certificate_key /path/to/key.pem;
```

```
location / {
    proxy_pass http://localhost:3000;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_set_header Host $host;
    proxy_cache_bypass $http_upgrade;
}

# Gzip compression
gzip on;
gzip_types text/plain text/css application/json application/javascript text/xml application/xml;
}
```

Environment Variables

Create .env file in production:

```
# Required for AI Chat feature
DEEPSEEK_API_KEY=sk-05bc8a86ded1487b9930f9637d6354b2

# Security
ALLOWED_ORIGINS=https://yourdomain.com,https://www.yourdomain.com

# Server
PORT=3000
```

SECURITY WARNING:

- NEVER commit .env to Git
- NEVER expose API keys in client-side code
- The provided key should ONLY be used server-side via the proxy

Google AdSense Setup

Before Applying (Prerequisites)

1. **Custom Domain:** Must use your own domain (not subdomain of free host)
2. **HTTPS:** SSL certificate required
3. **Content Depth:** Add 5-10 blog posts (920+ words each)
4. **Legal Pages:** Create Privacy Policy, Terms, Disclaimer pages
5. **Navigation:** All pages accessible from homepage
6. **No Placeholder Content:** Replace all “Lorem ipsum” text
7. **Mobile Responsive:** Test on multiple devices

Ad Unit Placement

Replace the placeholder `<div class="ad-slot">` elements with actual AdSense code:

```
<!-- Example: Top Ad Unit -->
<div id="ad-slot-top">
  <script async src="https://pagead2.googlesyndication.com/pagead/js/adsbygoogle.js?
    ↪ client=ca-pub-XXXXXXXXXXXXXXXXX"
    crossorigin="anonymous"></script>
  <ins class="adsbygoogle"
    style="display:block"
    data-ad-client="ca-pub-XXXXXXXXXXXXXXXXX"
    data-ad-slot="XXXXXXXXXX"
    data-ad-format="auto"
    data-full-width-responsive="true"></ins>
  <script>(adsbygoogle = window.adsbygoogle || []).push({});</script>
</div>
```

Ad Placement Strategy:

- **Top Banner:** Below hero section (high viewability)
- **Sidebar:** 300x250 or responsive (high CTR)
- **In-Content:** Between calculator inputs and results
- **Bottom:** After main content

AdSense Policy Compliance:

- Maximum 3 ads per page
- No ads near interactive elements (buttons, forms)
- No deceptive placement
- Content must be original and substantial
- No auto-refresh or pop-under ads

🔍 SEO Optimization Checklist

Pre-Launch

- Replace yourdomain.com in all meta tags with actual domain
- Update canonical URL in `<head>`
- Update Open Graph and Twitter card URLs
- Update `og:image` with actual branded image (1200x630px)
- Verify `robots.txt` points to correct sitemap URL
- Submit sitemap to Google Search Console
- Verify site ownership in Search Console
- Add Google Analytics 4 tracking code

Structured Data Verification

Test at: <https://search.google.com/test/rich-results>

PageSpeed Insights Targets

- Mobile LCP < 2.5s
- Mobile FID < 100ms
- Mobile CLS < 0.1
- Desktop score > 90

Content Strategy (Post-Launch)

- Publish 5 blog posts targeting long-tail keywords
 - Example: “how to calculate loan interest manually”
 - Example: “mortgage payoff calculator strategies”
 - Internal linking between blog posts and calculators
 - Add FAQ schema to popular calculators
-

Security Checklist

- Enable HTTPS (Let's Encrypt free SSL)
 - Set secure CORS headers in server.js
 - Add rate limiting (implemented in server.js)
 - Validate all API inputs
 - Set Content Security Policy headers
 - Enable HSTS
 - Add X-Frame-Options: DENY
 - Add X-Content-Type-Options: nosniff
-

PWA Configuration (Optional Enhancement)

Create manifest.json:

```
{
  "name": "FinanceTools Pro",
  "short_name": "FinanceTools",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#2563eb",
  "icons": [
    { "src": "/icon-192.png", "sizes": "192x192" },
    { "src": "/icon-512.png", "sizes": "512x512" }
  ]
}
```

Maintenance

Weekly

- Check Google Search Console for crawl errors
- Monitor Core Web Vitals
- Review AdSense performance reports

Monthly

- Update blog with 1-2 new posts
- Check for broken links
- Review and update calculator formulas
- Monitor server logs for errors

Quarterly

- SEO audit with SEMrush/Ahrefs
 - Update structured data if needed
 - Refresh content on top pages
 - Review ad placement performance
-

Troubleshooting

AdSense ads not showing?

1. Domain must be approved by AdSense (takes 1-2 weeks)
2. Check `ads.txt` file exists with: `google.com, pub-XXXXXXXXXXXXXXXX, DIRECT, f08c47fec0942fa0`
3. Ensure no ad blockers are active
4. Wait 48 hours after placing code

AI Chat not working?

1. Check `DEEPSEEK_API_KEY` is set in `.env`
2. Verify server is running: `curl http://localhost:3000/api/health`
3. Check browser console for CORS errors
4. Verify `ALLOWED_ORIGINS` includes your domain

SEO issues?

1. Test with Google Rich Results Test
 2. Check mobile-friendliness
 3. Verify `noindex` tags aren't present
 4. Ensure JavaScript renders correctly (use URL Inspection tool)
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Support

For issues or questions:

- Check server logs: `pm2 logs financetools`

- Test API: `curl -X POST http://localhost:3000/api/chat -H "Content-Type: application/json" -d '{"message": "test"}'`
- Review AdSense Help Center: <https://support.google.com/adsense>

Ready to deploy! 🚀